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Design and Pilot Validation of a Low-Cost Arduino-Based IoMT Platform for Multisensor Vital Sign Monitoring

Valeska Gatica-Rojas¹, Cristian Vidal-Silva²

¹ Human Motor Control Laboratory, Faculty of Health Sciences, Universidad de Talca. Av. Lircay S/N, Talca, 3460000, Chile; vgatica@utalca.cl

² Department of Interactive Visualization and Virtual Reality, Faculty of Engineering, Universidad de Talca. Av. Lircay S/N, Talca, Chile; cvidal@utalca.cl

* Correspondence: (V.G.-R.)

Abstract: Continuous monitoring of physiological variables is a relevant requirement in telemonitoring, telerehabilitation, and distributed healthcare scenarios. However, the integration of multiple sensing modules into compact, low-cost, and wireless platforms still involves important trade-offs related to communication overhead, energy autonomy, signal synchronization, and measurement reliability. This article presents the design, implementation, and preliminary validation of a low-cost Internet of Medical Things (IoMT) platform for multisensor vital sign monitoring based on the Arduino ecosystem and the Beetle ESP32-C6 V1.1 board. The proposed prototype integrates a MAX30102 module for heart rate and peripheral oxygen saturation monitoring, together with an SGP40-based exploratory module for respiratory-rate estimation. The system adopts a hierarchical data aggregation strategy in which sensor measurements are locally organized before transmission to the central processing unit, reducing redundant data exchange and supporting structured real-time communication. The prototype was evaluated in a laboratory setting using 20 assessments in healthy young adults under resting and mild dynamic conditions. Agreement with reference instruments was analyzed using correlation, mean bias, Bland–Altman analysis, and intraclass correlation coefficients. The results showed high agreement for heart rate and oxygen saturation, while respiratory-rate estimation showed promising but preliminary behavior that requires further validation with established respiratory sensors and broader populations. In addition, the paper reports the planned reproducibility package, including firmware, schematics, bill of materials, analysis scripts, and anonymized datasets. The findings suggest that the proposed architecture is a feasible engineering solution for low-cost multisensor telemonitoring prototypes, while its clinical use requires additional validation, regulatory assessment, and testing in more diverse physiological conditions.

Keywords: Internet of Medical Things; Arduino; ESP32-C6; multisensor monitoring; heart rate; oxygen saturation; respiratory-rate estimation; telemonitoring

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1. Introduction

Continuous monitoring of physiological variables has become increasingly relevant in telemonitoring, telerehabilitation, and distributed healthcare services [1–3]. Physiological parameters such as heart rate (HR), peripheral oxygen saturation (SpO_2), and respiratory rate (RR) provide valuable information for observing functional status, recovery progression, and physiological responses during daily activities and rehabilitation interventions [4]. In parallel, the rapid evolution of the Internet of Medical Things (IoMT) has enabled the development of wearable and portable monitoring platforms capable of acquiring physiological data outside traditional clinical environments [5–10].

Recent advances in wearable sensing technologies have significantly improved the accessibility of physiological monitoring systems, enabling continuous health assessment through compact and increasingly accurate wearable devices [11,12]. However,

important challenges remain regarding communication efficiency, energy autonomy, synchronization of heterogeneous sensor streams, and real-time operation in resource-constrained embedded platforms [11,13]. The integration of multiple physiological sensors into a compact wireless architecture requires balancing several engineering constraints, including sampling frequency, packet transmission overhead, processing capability, energy consumption, and communication latency [14–16].

Most existing low-cost monitoring platforms adopt parallel communication schemes in which individual sensing modules independently transmit data to a central processing unit. Although this approach simplifies system design, it may increase communication overhead, packet redundancy, synchronization complexity, and energy demand, particularly when multiple physiological signals are acquired simultaneously [17]. Furthermore, many embedded monitoring solutions rely on external infrastructure for processing and synchronization, limiting their suitability for autonomous operation in portable and wearable applications [18–20].

To provide a concise overview of the main challenges reported in recent IoMT monitoring systems, Table 1 summarizes the engineering limitations identified in the literature and the corresponding research direction adopted in this study.

Table 1: Research Motivation and Engineering Challenges Addressed in This Study

Dimension	Reported Limitation	Research Direction
Multisensor Integration	Independent acquisition modules often increase synchronization complexity.	Hierarchical organization of physiological measurements before transmission.
Communication Efficiency	Parallel architectures may generate redundant packet transmission.	Structured data aggregation strategy.
Embedded Processing	Resource-constrained platforms frequently depend on external processing infrastructure.	Local processing and packet organization within the embedded architecture.
Reproducibility	Many prototype-oriented studies provide limited implementation details.	Reproducible hardware–software design and documentation.
Scalability	Adding new sensing modules often increases communication complexity.	Modular architecture supporting heterogeneous sensing nodes.

Based on these challenges, this work presents a low-cost IoMT monitoring platform built upon the Arduino ecosystem and the Beetle ESP32-C6 V1.1 development board. The proposed architecture integrates a MAX30102 module for the acquisition of heart rate and oxygen saturation, together with an SGP40-based module for the evaluation of exploratory respiratory-rate. Unlike traditional parallel acquisition approaches, the system adopts a hierarchical communication strategy in which measurements are locally organized before transmission to the central processing unit.

Figure 1 illustrates the conceptual position of the proposed architecture within the current landscape of IoMT. Existing solutions typically focus on isolated aspects of the problem, such as wearable sensing, communication efficiency, or embedded implementation. The present study aims to integrate these dimensions into a unified and reproducible architecture suitable for low-cost physiological monitoring research.

The primary objective of this work is not to establish clinical deployment readiness, but rather to design, implement, and evaluate a low-cost engineering platform capable of supporting multisensor physiological monitoring under controlled laboratory conditions. Particular attention is devoted to communication organization, operational performance, reproducibility, and preliminary agreement analysis against reference measurements.

Despite the increasing availability of low-cost IoMT platforms, quantitative evidence regarding the impact of communication organization strategies remains limited. Most studies report physiological monitoring performance but rarely evaluate how packet aggregation influences latency, synchronization overhead, transmission efficiency, and energy consumption in embedded multisensor environments [18,19,21]. Furthermore, although recent studies have explored sensor orchestration and multisensor software architectures, the practical benefits of hierarchical communication architectures remain insufficiently documented in the current literature, particularly in low-cost embedded healthcare systems [16,17].

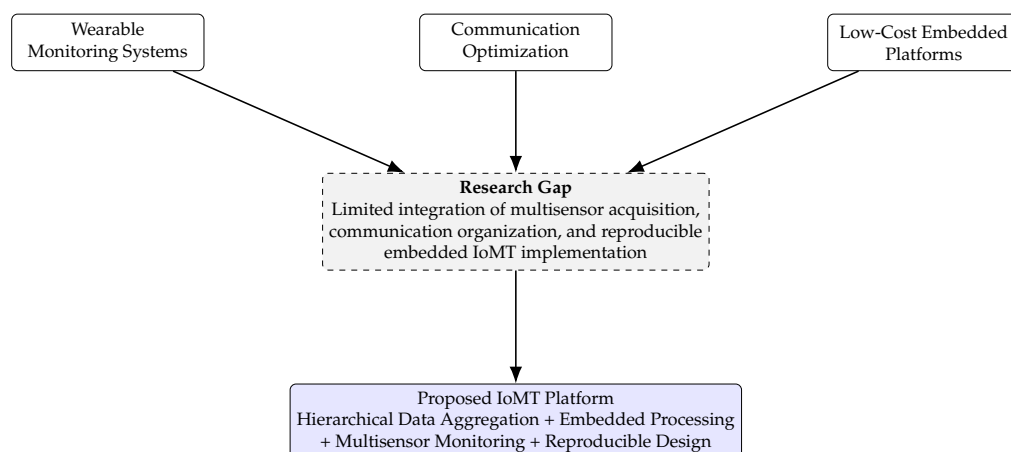


Figure 1. Conceptual positioning of the proposed platform. Existing studies commonly address wearable sensing, communication optimization, or embedded implementation independently. This work integrates these dimensions into a unified low-cost IoMT architecture for multisensor physiological monitoring.

1.1. Research Objective and Contributions

The increasing demand for portable physiological monitoring systems highlights the need for embedded architectures capable of integrating heterogeneous sensors while maintaining acceptable communication efficiency and operational performance. In response to this challenge, this study investigates whether a hierarchical communication strategy can provide practical advantages over conventional parallel transmission approaches in a low-cost IoMT environment.

Accordingly, the following research questions are addressed:

- RQ1** [Communication Organization]: How can a hierarchical data aggregation strategy improve the organization of heterogeneous physiological measurements within a low-cost embedded monitoring platform?
- RQ2** [Operational Performance]: What communication latency, packet-transmission behavior, and energy-related characteristics can be achieved using the Beetle ESP32-C6 architecture?
- RQ3** [Preliminary Measurement Agreement]: What level of agreement is observed between the proposed prototype and reference measurements under resting and mild dynamic conditions?

The main contributions of this article are summarized as follows:

- Development of a low-cost Arduino-compatible IoMT platform based on the Beetle ESP32-C6 V1.1 for integrated acquisition of heart rate, oxygen saturation, and exploratory respiratory-rate information.
- Design of a hierarchical communication architecture that organizes physiological measurements before wireless transmission through structured aggregation mechanisms.
- Quantitative evaluation of communication performance through packet transmission, latency analysis, and operational behavior assessment.
- Preliminary agreement analysis between the proposed platform and reference measurements obtained under controlled laboratory conditions.
- Reproducible hardware–software implementation framework including firmware, hardware configuration, signal-processing procedures, and experimental protocol documentation.

The remainder of this article is organized as follows. Section 2 reviews the literature on IoMT monitoring platforms, wearable sensing systems, embedded monitoring solutions, and communication architectures. Section 3 presents the hardware platform, hierarchical architecture, software environment, and communication framework adopted in the proposed monitoring system. Section 4 describes the experimental methodology, respiratory-rate estimation procedure, validation protocol, technical evaluation metrics, and statistical analysis. Section 5 reports the technical performance and physiological measurement agreement results obtained during laboratory evaluation. Section 6 interprets the findings, compares them with previous studies, discusses practical implications for IoMT-based telemonitoring systems, and examines the main deployment considerations and study limitations. Finally, Section 7 concludes the article and outlines future research directions.

2. Related Work

The development of Internet of Things (IoT) and Internet of Medical Things (IoMT) technologies has enabled new forms of real-time physiological monitoring based on wearable sensors, embedded platforms, wireless communication, and cloud or edge-based data processing [6,7,20]. These systems are increasingly relevant in telemonitoring, telerehabilitation, and decentralized healthcare scenarios, where physiological information must be acquired continuously and transmitted reliably outside conventional clinical environments. However, the design of multisensor IoMT platforms remains challenging. Physiological monitoring systems must coordinate heterogeneous data streams, maintain low communication latency, reduce energy consumption, and provide sufficient measurement reliability under static and dynamic conditions [14–16]. In low-cost embedded platforms, these requirements are particularly demanding because processing capacity, memory, wireless bandwidth, and battery autonomy are limited.

2.1. Real-Time IoT Communication and Orchestration

Real-time monitoring systems require communication mechanisms capable of supporting continuous data transmission with predictable latency. Traditional IoT architectures frequently rely on asynchronous communication models, which may introduce delays due to buffering, scheduling, retransmission, and network congestion [14]. In healthcare-related applications, latency and reliability become especially important because physiological data streams may support timely feedback, remote observation, and rehabilitation monitoring.

Recent studies have explored deterministic and latency-aware communication models. Time-sensitive networking has been proposed to support real-time telemedicine and e-health services by improving predictability in data transmission [22]. Similarly, ultra-reliable low-latency communication models have been studied for healthcare IoT environments, particularly in beyond-5G and 6G scenarios [13]. Nevertheless, these approaches usually emphasize network-level communication and do not always address the complete integration of embedded sensing, local aggregation, and multisensor physiological data handling.

2.2. Wearable Multisensor Systems and Sensor Fusion

Wearable devices have become central components of modern physiological monitoring systems [12,23–26]. They enable continuous acquisition of variables such as heart rate, oxygen saturation, respiratory activity, movement, and other biosignals. Multisensor configurations can improve robustness by combining heterogeneous sources of information and by compensating for the limitations of individual sensors [26].

Sensor fusion strategies are commonly implemented at signal, feature, or decision level to improve reliability and interpretation [16]. However, wearable systems must still address several practical difficulties, including motion artifacts, signal noise, skin–sensor contact instability, and synchronization between heterogeneous data streams [27,28].

These limitations are particularly relevant when monitoring physiological variables during movement or rehabilitation-related activities.

Several validation studies have reported promising results for wearable physiological monitoring in controlled or semi-controlled environments [4,29,30]. Even so, many systems still depend on external processing infrastructure, proprietary devices, or cloud-based services, which may limit local responsiveness and increase implementation complexity [3,11].

2.3. Embedded and Low-Cost Monitoring Platforms & Comparison of Representative Approaches

Low-cost embedded platforms, including Arduino-compatible and ESP32-based systems, have been widely used for accessible IoT and healthcare monitoring prototypes [21,31]. These platforms support rapid prototyping, broad sensor compatibility, and flexible software development, making them attractive for research, teaching, and early-stage biomedical engineering applications.

Open-source hardware and software have also contributed to the development of affordable healthcare technologies, especially in resource-constrained environments [32,33]. In addition, embedded physiological monitoring systems have been proposed for portable and multichannel acquisition tasks [18]. However, low-cost platforms frequently involve trade-offs related to computational capacity, memory availability, power consumption, and communication stability. As a result, some prototypes rely on simplified processing pipelines or external systems for data organization and analysis [19].

Table 2 compares representative studies according to system type, sensor handling, real-time support, embedded deployment, reproducibility orientation, and main limitations. This comparison shows that previous works often focus on one dimension of the problem, such as communication, wearable sensing, or embedded implementation, while fewer studies address these aspects within a unified low-cost IoMT architecture.

2.4. Research Gap

The reviewed literature suggests three main gaps. First, communication-oriented studies provide valuable advances in latency and reliability, but they often do not address the complete embedded multisensor acquisition pipeline. Second, wearable multisensor systems improve physiological data acquisition, but many depend on external processing infrastructure or proprietary architectures. Third, low-cost embedded platforms are accessible and flexible, but their use in structured, reproducible, and latency-aware multisensor IoMT architectures remains insufficiently explored.

Figure 2 summarizes the conceptual taxonomy derived from the related work. The proposed study is positioned at the intersection of low-cost embedded platforms, multisensor physiological acquisition, hierarchical communication, and reproducible IoMT implementation.

In response to these gaps, this work proposes a low-cost hierarchical multisensor IoMT platform that integrates embedded physiological acquisition, structured data aggregation, communication performance evaluation, and preliminary agreement analysis. The objective is not to replace certified clinical monitoring systems, but to provide a reproducible engineering framework for future telemonitoring and telerehabilitation research.

3. System Architecture and Hardware Platform

The proposed monitoring platform was designed as a low-cost Internet of Medical Things (IoMT) architecture capable of integrating multiple physiological sensing modules within a unified embedded framework. The design emphasizes modularity, communication efficiency, portability, and reproducibility while maintaining compatibility with open-source hardware and software technologies. Open-source development paradigms have become increasingly important for accelerating innovation, reducing

Table 2: Comparison of Representative Approaches in IoMT Communication, Wearable Sensing, and Embedded Monitoring Systems

Reference	System Type	Sensor Handling	Real-Time Support	Embedded Deployment	Reproducibility Focus	Main Limitation
Lu et al. [22]	Real-time communication	Network-level data streams	High	Partial	Limited	Focuses on communication infrastructure rather than embedded multisensor integration.
Toptsis et al. [16]	Wearable multisensor systems	Sensor fusion strategies	Partial	Yes	Moderate	Emphasizes software architecture and fusion, with limited focus on low-cost physiological prototypes.
Vahdati et al. [11]	Healthcare wearable systems	Physiological sensing	Limited	No	Limited	Many systems depend on external infrastructure and heterogeneous validation protocols.
Tsebesebe et al. [21]	Arduino-based systems	Basic sensor integration	Limited	Yes	Moderate	Low-cost devices are reviewed, but real-time prioritization and hierarchical aggregation remain underexplored.
Hasan et al. [18]	IoT physiological monitoring	Multichannel sensing	Partial	Yes	Moderate	Provides portable monitoring but with limited emphasis on hierarchical communication and scalability.
Uddin and Koo [19]	Remote patient monitoring	Biosensor and cloud connectivity	Partial	Partial	Limited	Cloud dependence may increase complexity and reduce local responsiveness.
This work	Low-cost hierarchical IoMT platform	HR, SpO_2 , and exploratory RR estimation	Yes	Yes	Yes	Integrates embedded sensing, hierarchical aggregation, technical evaluation, and reproducible implementation.

implementation costs, and promoting technology transfer in healthcare-oriented engineering projects [34,35].

Unlike conventional monitoring systems that rely on independent sensing modules and external processing infrastructures, the proposed platform adopts a hierarchical communication strategy in which physiological measurements are locally organized before transmission. This approach aims to reduce communication redundancy, simplify synchronization, and improve data organization during continuous monitoring tasks.

3.1. Hardware Platform

The embedded implementation is based on the Beetle ESP32-C6 V1.1 development board, selected because of its compact dimensions, low-power operation, integrated wireless connectivity, and support for modern IoT communication protocols. Similar embedded-development approaches have been widely adopted within the Arduino ecosystem due to their flexibility, accessibility, and suitability for multidisciplinary engineering applications [34–36].

The sensing subsystem integrates two primary modules. Heart rate and peripheral oxygen saturation measurements are acquired through the MAX30102 photoplethysmography sensor. Photoplethysmography has become one of the most widely adopted approaches for wearable cardiovascular monitoring because it enables compact, low-cost, and non-invasive physiological assessment [37,38]. Respiratory-rate estimation is explored using an SGP40 module positioned near the breathing flow to detect periodic signal variations associated with inhalation and exhalation cycles. Alternative respiratory-monitoring approaches based on indirect sensing mechanisms have recently attracted attention because of their potential for integration into wearable and telemonitoring environments [38,39].

Table 3 summarizes the principal hardware elements incorporated into the proposed platform.

Figure 3 shows the general hardware organization of the proposed platform. The Beetle ESP32-C6 operates as the central embedded unit, receiving physiological mea-

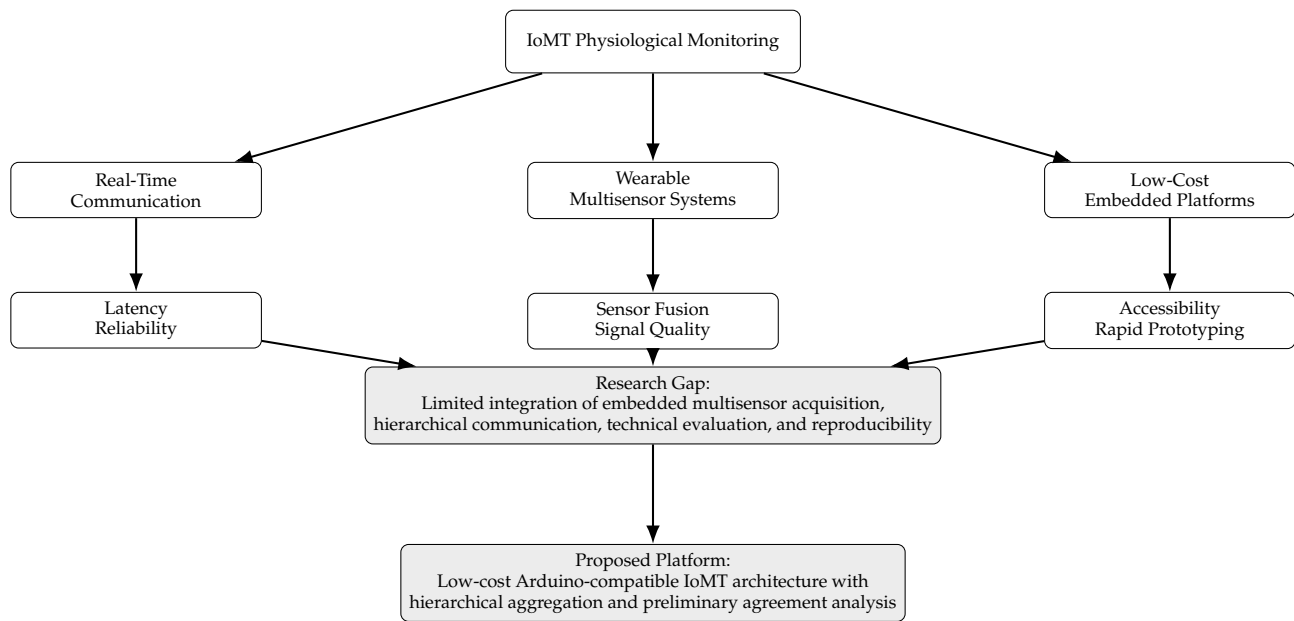


Figure 2. Taxonomy of related work and positioning of the proposed study. Existing approaches commonly emphasize communication, wearable sensing, or embedded implementation separately, whereas this work integrates these dimensions within a low-cost hierarchical IoMT architecture.

Table 3: Hardware Components of the Proposed Monitoring Platform

Component	Category	Function in the Platform
Beetle ESP32-C6 V1.1	Embedded controller	Coordinates sensing, local aggregation, packet construction, timing, and wireless communication.
MAX30102	Physiological sensor	Acquires photoplethysmographic signals for heart rate and peripheral oxygen saturation monitoring.
SGP40	Environmental sensor	Used as an exploratory indirect module for respiratory-rate estimation through breath-related signal fluctuations.
Battery module	Power subsystem	Provides portable operation during laboratory testing and supports future wearable deployment.
Wireless interface	Communication subsystem	Enables BLE/Wi-Fi communication and structured data transmission to the monitoring interface.
Monitoring interface	Application layer	Receives, visualizes, stores, and supports offline analysis of physiological measurements.

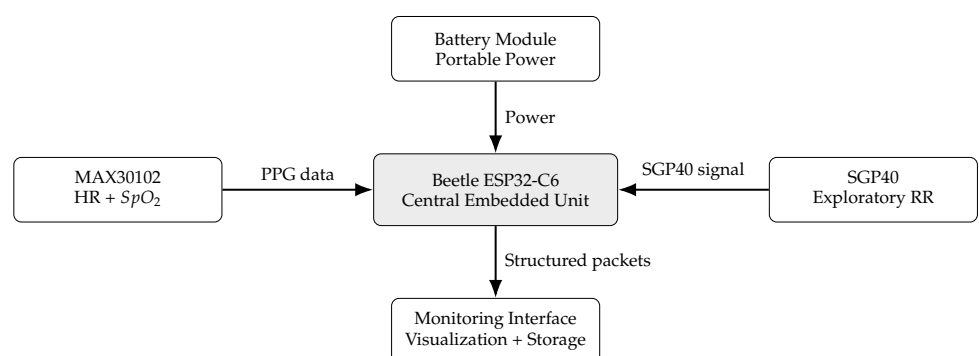


Figure 3. Hardware organization of the proposed monitoring platform. Physiological measurements are acquired through dedicated sensing modules and organized by the Beetle ESP32-C6 central unit before transmission to the monitoring interface.

surements from the sensing modules and forwarding structured information to the monitoring interface.

Table 4: Software Components and Communication Roles

Component	Role	Purpose in the Proposed Platform
BLEDevice.h / BLEScan.h	BLE management	Detection, connection, and subscription to BLE-compatible sensing services.
WiFi.h	Wireless connectivity	Network communication between the embedded platform and monitoring interface.
WiFiManager.h	Network provisioning	Simplifies configuration of wireless credentials and local network access.
ArduinoWebsockets.h	Data transmission	Supports bidirectional communication between the embedded unit and the monitoring layer.
Arduino_JSON.h	Data structuring	Formats physiological measurements into structured interoperable packets.
time.h / NTP	Synchronization	Provides timestamps for physiological samples and packet-level organization.
SSL/TLS certificates	Communication security	Supports encrypted transmission and server identity verification.

3.2. Hierarchical Multisensor Architecture

The proposed architecture is organized into four functional layers: sensing, aggregation, communication, and application. The sensing layer captures physiological information from dedicated modules. The aggregation layer locally organizes the acquired measurements and prepares structured packets. The communication layer transmits the resulting data through wireless mechanisms, while the application layer supports visualization, storage, and offline analysis.

This layered design differs from a conventional parallel architecture, in which each sensor transmits independently to the monitoring interface. In the proposed scheme, physiological measurements are organized before transmission, reducing redundant data exchange and supporting a clearer temporal structure for multisensor acquisition.

Figure 4 illustrates the hierarchical multisensor architecture adopted in this work. This figure reuses the previous architectural design of the manuscript, but it is now interpreted as an engineering-oriented IoMT architecture rather than as evidence of clinical deployment readiness.

3.3. Software Environment and Communication Framework

The software architecture was implemented using the Arduino development environment and libraries compatible with the ESP32 ecosystem. The implementation focuses on modular acquisition, structured data representation, timestamping, and wireless transmission. Instead of relying on a purely sensor-to-server scheme, the firmware organizes physiological data locally before transmission.

The main software components are summarized in Table 4. These components support wireless connectivity, packet construction, synchronization, and monitoring interface communication.

The communication process begins with sensor initialization and service detection. Once the sensing modules are available, the embedded controller acquires physiological measurements, applies basic preprocessing and packet formatting, and transmits the structured data to the monitoring interface. Each packet includes physiological measurements, temporal information, sensor identification, and acquisition-status indicators.

The detailed structure of the physiological packet and its role within the hierarchical communication framework are described in Section 3.4.

3.4. Physiological Packet Structure and Communication Organization

To facilitate synchronization, interoperability, and efficient wireless transmission, the proposed platform organizes all physiological measurements into a unified packet structure before transmission. Rather than transmitting each sensing stream independently, the embedded controller aggregates the available measurements and generates a structured packet containing physiological values, temporal information, and acquisition-status indicators.

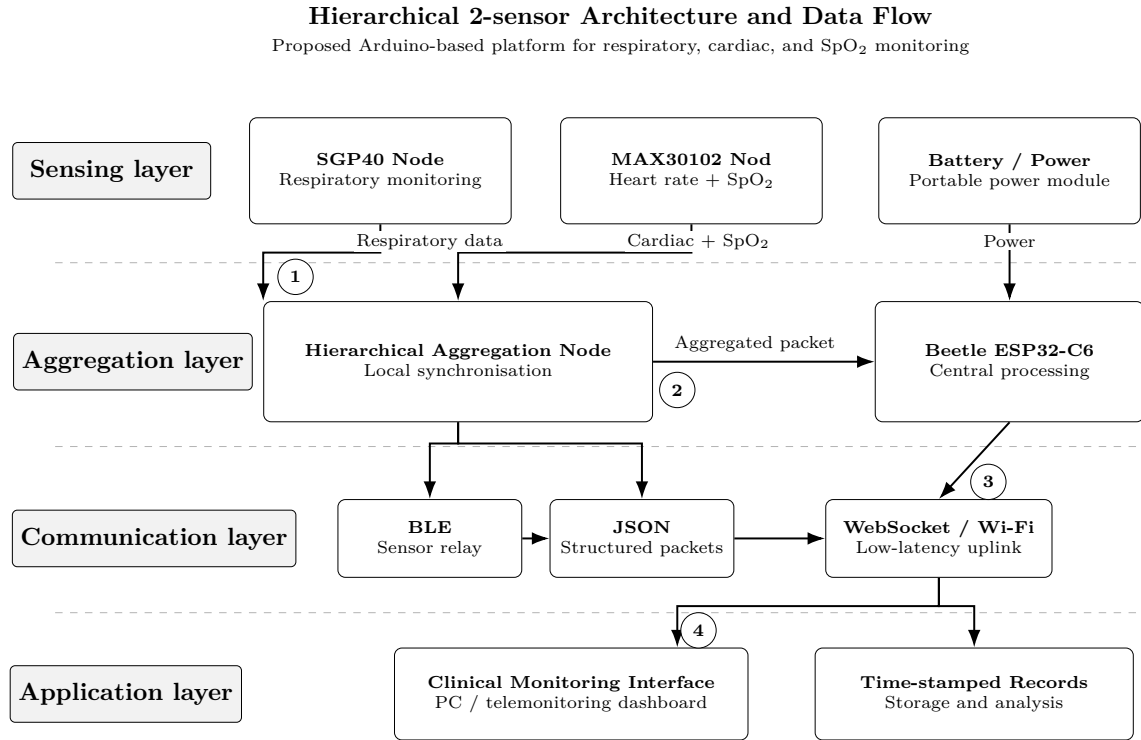


Figure 4. Hierarchical multisensor architecture adopted in the proposed IoMT platform. Physiological measurements are acquired by dedicated sensing nodes, locally aggregated, and subsequently transmitted through a structured communication framework to the monitoring layer.

Node ID	Timestamp	HR	SpO ₂	RR	Status
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Figure 5. Logical organization of the physiological packet transmitted by the proposed monitoring platform. Each packet contains identification, synchronization, physiological measurements, and acquisition-status information.

The generic packet representation adopted in this study is defined as:

$$P_i = \{ID_s, t_i, HR_i, SpO_{2i}, RR_i, q_i\}, \quad (1)$$

where P_i denotes the packet generated at acquisition instant i , ID_s represents the sensor or node identifier, t_i corresponds to the timestamp, HR_i is the heart-rate measurement, SpO_{2i} is the oxygen-saturation value, RR_i is the respiratory-rate estimate, and q_i represents a quality or acquisition-status indicator.

This packet organization enables all physiological variables to be synchronized through a common timestamp while reducing communication complexity and facilitating subsequent analysis. Furthermore, the structure is extensible and allows additional sensing modalities to be incorporated without modifying the overall communication framework.

Figure 5 illustrates the logical organization of the physiological packet transmitted by the proposed platform.

The adoption of a unified packet structure improves traceability during physiological acquisition, simplifies storage and post-processing procedures, and supports reproducible analysis workflows. Additionally, the approach provides a scalable foun-

Table 5: Planned Repository Structure for Reproducible Implementation

Folder	Content
/firmware	ESP32-C6 source code, sensor acquisition modules, communication routines, and packet construction logic.
/hardware	Circuit schematic, wiring diagram, bill of materials, and assembly documentation.
/analysis	Python or MATLAB scripts for preprocessing, latency analysis, agreement analysis, and visualization.
/data	Anonymized raw and processed measurements obtained during laboratory validation.
/docs	Protocol description, calibration notes, configuration instructions, and usage guidelines.

dition for future integration of inertial, environmental, or rehabilitation-specific sensing modules within the same communication architecture.

3.5. Reproducibility-Oriented Implementation

The platform was developed with a reproducibility-oriented structure. This design philosophy is consistent with current recommendations promoting open-source hardware, open software, and transparent development practices in biomedical engineering and healthcare technologies [32,33]. The reproducibility package associated with the implementation is organized into firmware, hardware documentation, analysis scripts, and experimental data. The intended repository structure is shown in Table 5.

By documenting both hardware and software components, the proposed platform aims to support independent verification and future comparative studies. This aspect is particularly relevant for low-cost IoMT prototypes, where reproducibility is often limited by incomplete reporting of firmware, circuit design, or data-processing procedures.

4. Experimental Methodology

The objective of the experimental methodology was to evaluate both the operational behavior of the proposed IoMT platform and the preliminary agreement of the physiological measurements obtained under controlled laboratory conditions. Unlike clinical validation studies designed to demonstrate diagnostic performance, the present work focuses on engineering feasibility, communication performance, and measurement agreement analysis.

The methodology was organized into four stages: (1) physiological signal acquisition, (2) local aggregation and packet generation, (3) wireless transmission and storage, and (4) technical and statistical evaluation.

Figure 6 summarizes the complete evaluation framework adopted in this study. The protocol combines simultaneous physiological acquisition, technical performance assessment, and statistical agreement analysis in order to evaluate both the engineering and physiological behavior of the proposed platform.

4.1. Participants

A controlled laboratory protocol was conducted at the Human Motor Control Laboratory. Twenty assessments were performed in healthy young adults between 20 and 25 years of age. Participants reported no known cardiovascular, respiratory, or neurological disorders at the time of evaluation. The present evaluation was conceived as an engineering-oriented feasibility and agreement study rather than a clinical validation trial. Therefore, the sample size was selected to assess technical performance and preliminary physiological agreement under controlled laboratory conditions.

Table 6 summarizes the demographic characteristics of the participant cohort included in the evaluation protocol.

4.2. Physiological Acquisition Modules

Three physiological variables were considered in the proposed platform: heart rate, peripheral oxygen saturation, and respiratory rate.

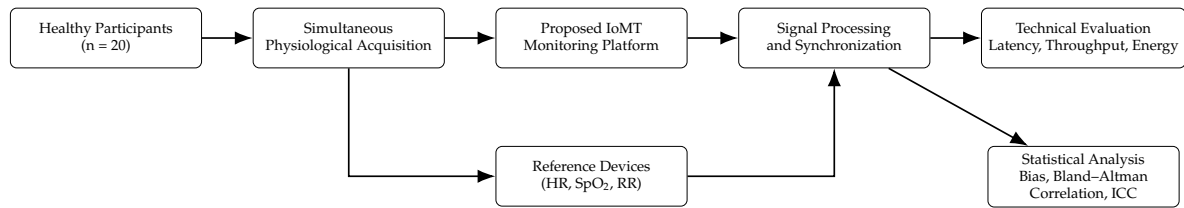


Figure 6. Experimental framework adopted for the evaluation of the proposed IoMT platform. Simultaneous physiological acquisition was performed using both the proposed system and reference measurements, followed by technical performance assessment and statistical agreement analysis.

Table 6: Participant Characteristics

Variable	Value
Participants	20
Age (years)	22.4 ± 1.8
Male	11
Female	9
Body Mass Index (kg/m^2)	24.1 ± 3.2

Table 7: Physiological Variables Included in the Monitoring Platform

Variable	Sensor	Purpose
Heart Rate	MAX30102	Continuous monitoring of cardiovascular activity.
SpO_2	MAX30102	Monitoring of peripheral oxygen saturation.
Respiratory Rate	SGP40	Exploratory estimation of respiratory periodicity through breath-related signal variation.

Heart rate and oxygen saturation were acquired using the MAX30102 photoplethysmography sensor. Photoplethysmographic sensing has become one of the most widely adopted approaches for wearable cardiovascular monitoring due to its low cost, compact implementation, and suitability for embedded healthcare systems [25,37].

Respiratory-rate estimation was explored using an SGP40 module positioned near the breathing flow. Although the SGP40 was not originally designed as a respiratory sensor, recent reviews have highlighted the growing interest in alternative sensing modalities and indirect respiratory estimation techniques for low-cost wearable monitoring systems [24,39]. Instead, the approach adopted in this work relies on breath-related fluctuations detected in the sensor output. Consequently, the respiratory module should be interpreted as an exploratory respiratory-rate estimator rather than a clinically validated respiratory sensing solution.

Table 7 summarizes the physiological variables considered in the study.

4.3. Respiratory-Rate Estimation Procedure

The respiratory-rate estimation process consisted of signal acquisition, noise filtering, baseline correction, peak identification, cycle detection, respiratory-rate estimation, and quality assessment. The objective was to identify periodic signal fluctuations associated with respiratory activity while reducing the influence of environmental noise, sensor drift, and motion-related disturbances commonly reported in wearable respiratory monitoring systems [24,27,28,39].

Figure 7 illustrates the signal-processing and decision workflow adopted for the evaluation of the exploratory respiratory-rate from the output of the SGP40 sensor. The workflow explicitly separates acquisition, preprocessing, cycle validation, quality assessment, and packet update before transmission to the monitoring layer. For the exploratory respiratory-rate estimation procedure, the SGP40 signal was analyzed using a sliding-window strategy with a window length of $T_w = 30$ s and an overlap of $O = 50\%$ between consecutive windows. This configuration was selected to include several respiratory cycles within each analysis window, considering that the normal adult respiratory rate generally ranges between approximately 12 and 20 breaths per minute

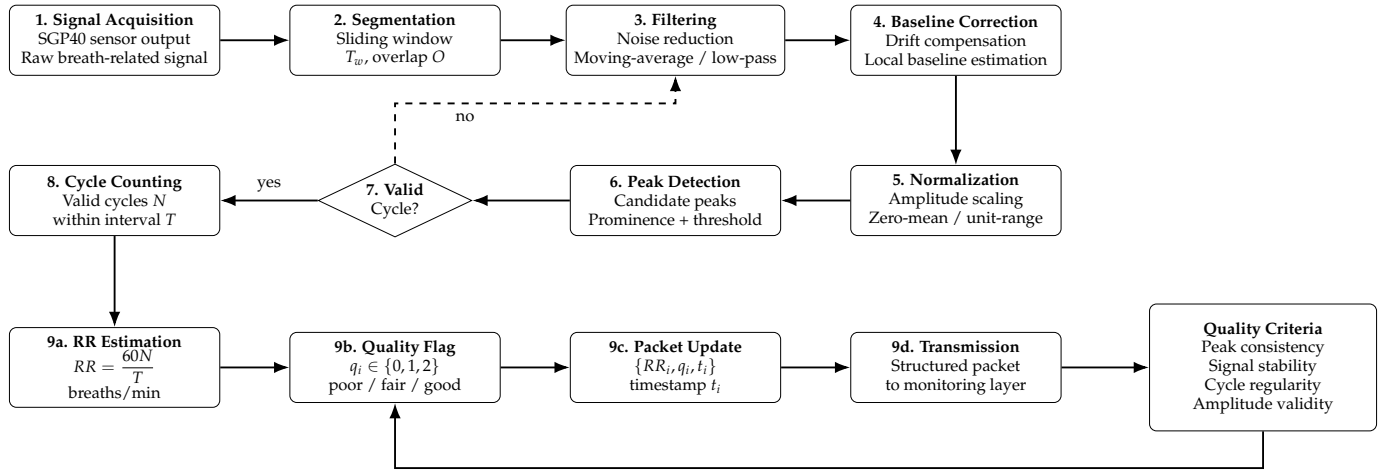


Figure 7. Signal-processing and decision workflow adopted for exploratory respiratory-rate estimation from the SGP40 sensor output. The procedure includes sliding-window segmentation, filtering, baseline correction, amplitude normalization, peak candidate detection, cycle validation, respiratory-rate computation, quality-flag assignment, packet updating, and transmission to the monitoring layer.

under resting conditions. Candidate peaks were accepted only when they satisfied the amplitude, temporal consistency, and cycle-regularity criteria.

The respiratory-rate estimation process consisted of three coordinated stages: pre-processing, cycle detection and validation, and output generation. The preprocessing stage reduced noise and compensated for baseline drift. The detection stage identified candidate respiratory cycles and rejected invalid cycles according to amplitude and temporal consistency criteria. Finally, the output stage computed respiratory rate, assigned a quality flag, updated the physiological packet, and transmitted the result to the monitoring layer.

4.4. Validation Protocol

The objective of the validation protocol was to compare physiological measurements generated by the proposed IoMT platform against reference measurements acquired simultaneously under equivalent experimental conditions.

The protocol consisted of two experimental phases:

1. **Resting condition:** participants remained seated for five minutes while continuous physiological measurements were acquired.
2. **Mild dynamic condition:** participants performed repeated squat movements to evaluate system behavior under motion and physiological activation.

Reference measurements were acquired simultaneously using validated external instruments in order to provide a benchmark for agreement analysis. Heart-rate measurements were compared against a Polar FT7 heart-rate monitor, oxygen-saturation measurements were compared against a clinical pulse oximeter, and respiratory-rate measurements were compared against manual respiratory counts performed by a trained observer. Simultaneous acquisition was performed to ensure temporal alignment between the proposed platform and the reference measurements.

To ensure temporal consistency, measurements obtained from the proposed platform and the reference instruments were synchronized using common acquisition intervals and subsequently paired for statistical agreement analysis.

Table 8 summarizes the reference instruments used during the validation protocol.

These reference measurements served as the basis for the agreement analyses presented in Section 5, including bias estimation, Bland–Altman analysis, Pearson correlation analysis, and Intraclass Correlation Coefficient (ICC) calculations.

Table 9 summarizes the experimental validation protocol.

Table 8: Reference Instruments Used During Validation

Variable	Reference Device	Purpose
Heart Rate	Polar FT7	Reference cardiovascular monitoring.
SpO ₂	Clinical pulse oximeter	Reference oxygen-saturation measurement.
Respiratory Rate	Manual respiratory count by trained observer	Reference respiratory-rate assessment.

Table 9: Experimental Validation Protocol

Phase	Duration	Objective
Resting condition	5 min	Baseline physiological acquisition under stable conditions.
Dynamic condition	5 min	Assessment during mild movement and physiological activation.

4.5. Technical Evaluation

In addition to physiological measurements, the communication architecture was evaluated using technical indicators associated with transmission performance and operational behavior. Latency was calculated from 100 consecutive packet transmissions recorded during continuous physiological acquisition. For each packet, timestamps were registered at generation and reception to estimate end-to-end communication delay.

Current consumption was measured under both active acquisition and standby operating modes. These measurements were subsequently used to estimate operational autonomy under the evaluated configuration. Table 10 summarizes the engineering metrics considered.

Table 10: Technical Performance Measurement Framework

Metric	Description	Unit
Packets per minute	Communication load generated by the platform	packets/min
Mean latency	Average end-to-end communication delay	ms
Latency percentiles	Worst-case communication behavior	ms
Active current draw	Current consumption during acquisition	mA
Standby current draw	Current consumption during idle operation	mA
Estimated autonomy	Projected operating time under the evaluated configuration	h

The technical evaluation focused on communication load, latency behavior, and energy consumption indicators, providing an engineering-oriented assessment of the proposed platform under continuous physiological acquisition conditions.

4.6. Statistical Analysis

The acquired data were exported for offline analysis using Python, MATLAB, and SPSS. Signal preprocessing included filtering procedures to reduce motion artifacts and high-frequency noise [27,28].

Agreement analysis was performed using Bland–Altman methodology [40], allowing estimation of mean bias and limits of agreement between the proposed platform and the reference measurements. For each physiological variable, the limits of agreement (LoA) were calculated as the mean difference between the proposed platform and the reference measurement plus and minus 1.96 standard deviations of the measurement differences.

Pearson correlation coefficients were computed to quantify linear association between measurements. However, because correlation alone does not necessarily imply measurement interchangeability, agreement-based statistics were prioritized throughout the analysis. In addition, Intraclass Correlation Coefficients (ICC) were calculated

Table 11: Technical Performance Indicators of the Proposed Platform

Metric	Value
Packets transmitted per minute	120
Mean latency (ms)	23.0
Latency SD (ms)	1.0
95th percentile latency (ms)	24.8
99th percentile latency (ms)	25.3
Active current draw (mA)	84.7
Standby current draw (mA)	11.8
Estimated autonomy (h)	22.4

to evaluate measurement consistency and agreement [41]. Statistical significance was established at $\alpha = 0.05$.

Because the participant cohort consisted exclusively of healthy young adults, particular attention was given to the interpretation of oxygen saturation measurements, which were expected to remain within a relatively narrow physiological range. Overall, the adopted methodology combines engineering-oriented performance indicators with physiological agreement analyses, providing a comprehensive framework for evaluating the proposed IoMT monitoring platform under controlled laboratory conditions.

5. Results

The proposed platform was evaluated from two complementary perspectives. First, the communication architecture was assessed through technical indicators associated with packet transmission, latency, and operational behavior. Second, physiological measurements obtained from the embedded sensing modules were compared against reference measurements collected under equivalent experimental conditions.

The results are therefore organized into technical performance indicators and physiological agreement analyses. This separation allows the engineering contribution of the platform to be evaluated independently from the measurement agreement obtained during laboratory testing.

5.1. Technical Performance Evaluation

The communication architecture was evaluated using packet transmission, latency, and operational behavior indicators. The objective was to determine whether the proposed hierarchical organization could support continuous physiological monitoring while maintaining acceptable responsiveness.

Table 11 summarizes the technical performance indicators obtained during the evaluation.

Estimated autonomy was calculated from the measured active current draw assuming a rechargeable Li-Po battery with an effective capacity of approximately 1900 mAh. Consequently, the reported autonomy should be interpreted as a theoretical operating estimate derived from the measured power consumption rather than as the operating lifetime of the prototype during the laboratory experiments.

Overall, the technical evaluation indicates that the proposed communication framework maintained stable transmission throughout the acquisition sessions. Latency remained consistently low, packet transmission was stable during continuous acquisition, and the measured energy consumption suggests that the platform is suitable for extended laboratory-based physiological monitoring. These indicators should be interpreted as engineering performance metrics rather than evidence of clinical deployment readiness.

Figure 8 shows the latency observed during 100 consecutive packet transmissions. The measurements remained within a narrow operating range, with all values below 30 ms and most observations concentrated between 22 and 25 ms. These results suggest stable communication behavior and indicate that the proposed hierarchical architec-

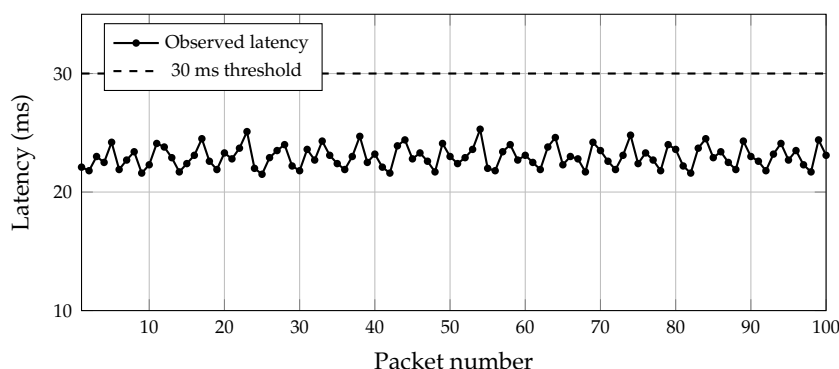


Figure 8. Observed latency distribution during continuous physiological monitoring. The proposed hierarchical communication strategy maintained stable transmission behavior, with latency values consistently below 30 ms throughout the acquisition sessions.

ture can support continuous physiological monitoring without introducing substantial transmission delays.

5.2. Physiological Measurement Agreement

The physiological performance of the proposed platform was evaluated by comparing heart rate (HR), peripheral oxygen saturation (SpO_2), and respiratory-rate (RR) measurements against their corresponding reference measurements under both static and dynamic conditions.

The main findings can be summarized as follows:

- **Heart Rate (HR):** Heart-rate measurements obtained from the MAX30102 module exhibited excellent agreement with the reference device. Under static conditions, the system achieved a Pearson correlation coefficient of 0.98 ($p < 0.001$), a mean bias of 1.2 bpm, and an ICC of 0.97 (95% CI: 0.95–0.99). During the dynamic protocol, agreement remained high despite the presence of motion artifacts, with a correlation coefficient of 0.94 ($p < 0.001$), a mean bias of 2.5 bpm, and an ICC of 0.93 (95% CI: 0.89–0.96). The observed reduction in agreement is consistent with the known sensitivity of photoplethysmographic sensors to movement and variations in sensor positioning.
- **Oxygen Saturation (SpO_2):** Oxygen-saturation measurements remained within the expected physiological range for healthy young adults. The module achieved correlation coefficients of 0.96 and 0.92 under static and dynamic conditions, respectively, while maintaining mean biases below 1.5%. Because most participants exhibited oxygen-saturation values between 95% and 100%, the observed variability was naturally limited. Consequently, the results primarily demonstrate measurement consistency under normoxic conditions rather than performance under clinically significant oxygen desaturation.
- **Respiratory Rate (RR):** The exploratory respiratory-rate estimation module successfully identified respiratory periodicity during both resting and mild dynamic activities. Under static conditions, respiratory-rate estimation achieved a correlation coefficient of 0.95 ($p < 0.001$), a mean bias of 0.8 breaths per minute, and an ICC of 0.94 (95% CI: 0.90–0.97). During dynamic evaluation, the correlation coefficient decreased to 0.91 ($p < 0.001$), while the mean bias increased to 1.9 breaths per minute and the ICC reached 0.89 (95% CI: 0.85–0.93). Although the respiratory-rate estimates followed the expected physiological trends, respiratory monitoring remains one of the most challenging physiological measurements in wearable environments due to motion artifacts, environmental variability, and sensor-placement effects [24,27,39].

Table 12 summarizes the agreement indicators obtained for all physiological variables. Overall, the results demonstrated high agreement across measurements, with stronger performance under static conditions and a modest reduction during the dynamic protocol. The Bland–Altman analysis (Figure 9) indicated acceptable agreement between the proposed platform and the reference measurements, with most observations lying within the corresponding limits of agreement. Consistent with the correlation and ICC results, heart-rate measurements exhibited the highest level of agreement, whereas respiratory-rate estimation showed greater variability, particularly under dynamic conditions. Nevertheless, the overall findings support the feasibility of the proposed monitoring platform for physiological data acquisition under controlled laboratory conditions.

Figure 10 complements Table 12 by illustrating the ICC values obtained for each physiological variable under static and dynamic conditions. Agreement remained high across all variables, although a modest reduction was observed during movement-related activities.

Taken together, these results indicate that the proposed platform provides reliable physiological measurements under controlled laboratory conditions. Heart rate and oxygen saturation exhibited the strongest agreement with reference measurements, while respiratory-rate estimation demonstrated promising behavior despite the exploratory nature of the sensing approach.

6. Discussion

6.1. Interpretation of Findings

The results obtained in this study demonstrate the feasibility of implementing a low-cost physiological monitoring platform using an embedded IoMT architecture based on the Beetle ESP32-C6 ecosystem. From an engineering perspective, the proposed platform successfully integrated heterogeneous sensing modules, local data aggregation mechanisms, and wireless communication services within a unified architecture capable of supporting continuous physiological monitoring. These findings are consistent with previous IoMT healthcare architectures that have demonstrated the feasibility of combining low-cost sensing, wireless communication, and distributed monitoring functionalities in clinical and home-care environments [9,10].

One of the most relevant findings concerns the communication performance of the hierarchical architecture. The technical evaluation revealed stable packet transmission and consistently low latency values throughout the acquisition sessions. Mean latency remained below 25 ms, while all observed transmissions remained below the predefined 30 ms threshold. These results suggest that the proposed communication strategy can support real-time physiological monitoring scenarios without introducing substantial transmission delays.

Regarding physiological measurements, heart rate exhibited the strongest agreement with the reference measurements. This finding is consistent with previous studies reporting the robustness of photoplethysmography-based sensing under controlled laboratory conditions [25,29,37]. Although a modest reduction in agreement was observed during movement, the resulting correlation and ICC values remained high, indicating that the platform was capable of capturing meaningful cardiovascular responses during both resting and mild dynamic activities.

During the mild dynamic condition, motion-related artifacts were expected to affect the MAX30102 photoplethysmographic signal due to changes in skin–sensor contact, local pressure, and acceleration-induced noise. To reduce these effects, the preprocessing stage incorporated basic filtering and rejection of unstable samples before agreement analysis. Nevertheless, motion artifacts remain a relevant limitation of photoplethysmography-based monitoring during physical activity. Future versions of the platform should integrate an inertial measurement unit (IMU) to support motion-aware artifact detection and active compensation during dynamic rehabilitation tasks.

Table 12: Summary of Agreement Indicators Obtained for the Physiological Variables

Variable	Condition	r	p-value	Mean Bias	ICC (95% CI)
Heart Rate	Static	0.98	< 0.001	1.2 bpm	0.97 (0.95–0.99)
	Dynamic	0.94	< 0.001	2.5 bpm	0.93 (0.89–0.96)
SpO ₂	Static	0.96	< 0.001	0.80%	0.95 (0.92–0.98)
	Dynamic	0.92	< 0.001	1.40%	0.91 (0.87–0.94)
Respiratory Rate	Static	0.95	< 0.001	0.8 brpm	0.94 (0.90–0.97)
	Dynamic	0.91	< 0.001	1.9 brpm	0.89 (0.85–0.93)

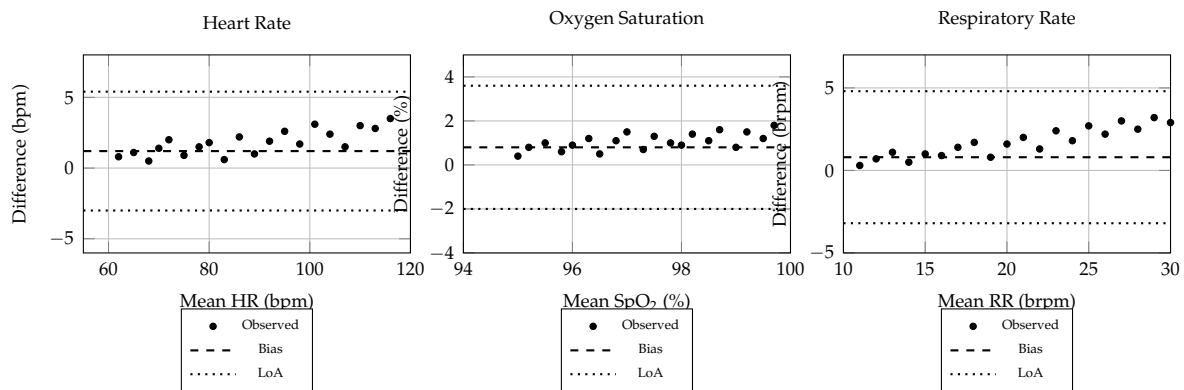


Figure 9. Bland–Altman agreement analysis for heart rate, oxygen saturation, and respiratory-rate estimation. Dashed lines represent mean bias, while dotted lines represent the approximate limits of agreement.

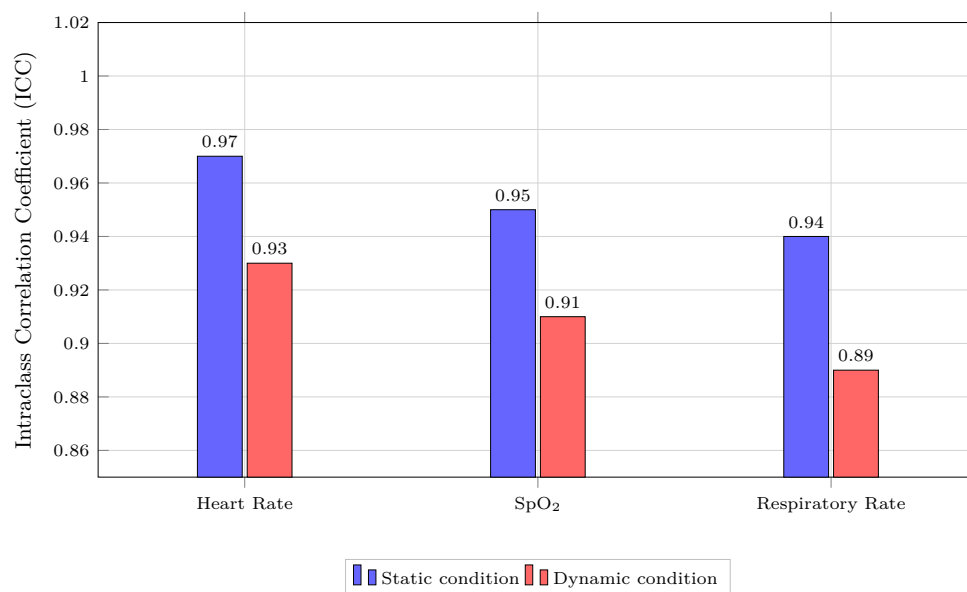


Figure 10. Intraclass correlation coefficients obtained for heart rate, oxygen saturation, and respiratory-rate estimation under static and dynamic conditions. Agreement remained high across variables, although a modest reduction was observed during movement-related activities.

The oxygen saturation measurements also demonstrated high consistency with the corresponding reference measurements. However, the relatively narrow physiological range observed among the participants must be considered when interpreting these results. Since all evaluations were performed in healthy young adults, the agreement indicators obtained primarily reflect measurement consistency under normoxic conditions rather than performance under clinically significant oxygen desaturation.

The estimation of Respiratory-rate produced encouraging results despite the exploratory nature of the adopted sensing approach. Although the SGP40 module was not originally designed as a respiratory sensor, the signal-processing pipeline successfully identified respiratory periodicity and generated estimates that followed expected physiological trends. Nevertheless, the observed reduction in agreement during movement suggests that respiratory estimation remains the most challenging component of the proposed platform and should be interpreted as a proof-of-concept rather than a clinically validated solution.

6.2. Comparison with Existing IoMT Monitoring Platforms

The findings of this study are consistent with the broader literature on low-cost physiological monitoring systems and IoMT architectures. Previous research has demonstrated the potential of wearable sensing technologies for acquiring cardiovascular and respiratory information in telemonitoring environments [11,16,23]. However, many reported solutions rely on proprietary hardware, external processing infrastructures, or cloud-dependent architectures that may increase deployment complexity and reduce local responsiveness.

In contrast, the platform proposed in this study emphasizes local processing, structured packet organization, and embedded implementation using open-source technologies. This design philosophy contributes to reproducibility and facilitates adaptation to diverse research and educational contexts.

The communication performance observed in this work is also consistent with recent efforts focused on low-latency IoMT systems [13,22]. Although the present study does not attempt to compete with specialized industrial communication frameworks such as Time-Sensitive Networking (TSN), the obtained latency values suggest that low-cost embedded platforms can achieve operational performance levels sufficient for many telemonitoring and rehabilitation applications.

Compared with previous Arduino-compatible monitoring systems [18,21,31], the proposed architecture additionally incorporates a hierarchical aggregation strategy that organizes physiological measurements before transmission. While the current study did not include a direct comparison against a parallel communication architecture, the results indicate that structured aggregation can provide a practical framework for managing heterogeneous physiological streams within resource-constrained embedded environments.

6.3. Practical Implications for Telemonitoring and IoMT Research

The proposed platform has several practical implications for future telemonitoring and IoMT research. First, the architecture demonstrates that low-cost embedded technologies can support simultaneous acquisition of multiple physiological variables without requiring complex external infrastructures. This characteristic may be particularly valuable in educational laboratories, research prototypes, and resource-constrained environments.

Second, the use of open-source hardware and software components facilitates reproducibility and technology transfer. Researchers can replicate, modify, and extend the platform without dependence on proprietary ecosystems, thereby accelerating experimentation and comparative evaluation.

Third, the hierarchical communication strategy provides a scalable foundation for integrating additional sensing modalities. Future implementations could incorporate inertial sensors, electromyography, environmental monitoring, or rehabilitation-specific sensors while preserving the same communication framework.

Finally, the architecture may serve as a useful experimental platform for investigating edge-computing strategies, local signal processing, and distributed IoMT deployments. These research directions are becoming increasingly relevant as healthcare technologies move toward decentralized and home-based monitoring environments.

6.4. Regulatory and Deployment Considerations

Although the obtained results are encouraging, the proposed platform should be considered a research prototype rather than a medical device intended for clinical deployment. The present study focused on engineering feasibility and preliminary measurement agreement under controlled laboratory conditions.

Before any clinical implementation, additional development stages would be required, including large-scale validation studies involving diverse patient populations, long-term reliability assessments, risk-management procedures, and compliance with applicable regulatory frameworks. Depending on the target jurisdiction and intended use, such processes may involve standards associated with electrical safety, electromagnetic compatibility, software lifecycle management, cybersecurity, and medical-device certification.

Consequently, the results reported in this work should not be interpreted as evidence of regulatory readiness. Instead, they provide a foundation for future investigations aimed at bridging the gap between low-cost IoMT research prototypes and clinically deployable physiological monitoring systems.

6.5. Study Limitations

Several limitations should be considered when interpreting the findings of this study:

- **Participant population.** The participant cohort was relatively small and consisted exclusively of healthy young adults. Consequently, the obtained agreement indicators may not generalize to elderly populations, pediatric populations, or individuals with cardiovascular, respiratory, or metabolic disorders.
- **Oxygen-saturation range.** Oxygen-saturation measurements were obtained within a relatively narrow physiological range. This limitation restricted the evaluation of the sensing module under clinically relevant hypoxemic conditions and may have contributed to the high agreement values observed.
- **Respiratory-rate estimation.** Respiratory-rate estimation relied on an exploratory methodology based on the SGP40 sensor. Although encouraging results were obtained, the sensor was not originally designed for respiratory monitoring, and additional validation against dedicated respiratory sensing technologies is required before considering broader healthcare applications.
- **Communication architecture comparison.** The present study did not include a direct experimental comparison between the proposed hierarchical communication architecture and an equivalent parallel communication architecture. Therefore, although the obtained latency and communication indicators suggest satisfactory performance, the quantitative benefit attributable specifically to hierarchical aggregation remains to be demonstrated experimentally.
- **Controlled laboratory conditions.** All evaluations were conducted in a controlled laboratory environment. Future studies should investigate system performance during longer monitoring periods and under real-world telemonitoring conditions, where environmental variability, wireless interference, network congestion, and user behavior may influence operational performance.

7. Conclusions and Future Work

This study presented a low-cost Internet of Medical Things (IoMT) platform based on the Beetle ESP32-C6 ecosystem for the integrated acquisition of heart rate, peripheral oxygen saturation, and exploratory respiratory-rate information. The proposed architecture combines embedded sensing, hierarchical data aggregation, and wireless communication within a unified framework designed to support physiological monitoring applications.

From an engineering perspective, the results demonstrated that the platform maintained stable communication performance throughout the evaluation protocol. The

hierarchical communication strategy achieved low-latency operation, consistent packet transmission, and satisfactory energy-related characteristics, indicating its suitability for real-time monitoring scenarios and future wearable implementations.

Regarding physiological measurements, the platform achieved high agreement with the reference measurements for heart rate and oxygen saturation under both static and dynamic conditions. Respiratory-rate estimation also showed promising results, successfully identifying respiratory periodicity and tracking expected physiological trends. Nevertheless, because the SGP40 sensor was employed as an exploratory respiratory sensing approach, the reported respiratory-rate results should be interpreted as preliminary rather than clinically validated.

The findings suggest that low-cost embedded architectures can support reliable multisensor physiological monitoring while maintaining accessibility, portability, and reproducibility. Furthermore, the use of open-source hardware and software components facilitates future adaptation, replication, and extension of the proposed platform in both research and educational environments.

Several opportunities for future work remain. First, larger validation studies involving diverse participant populations and clinically relevant physiological conditions are necessary to further assess measurement robustness. Second, direct comparisons between hierarchical and parallel communication architectures would provide a more rigorous evaluation of communication efficiency and energy consumption. Third, future developments could incorporate additional sensing modalities, edge-computing capabilities, and advanced signal-processing techniques to enhance system functionality. Finally, long-term evaluations in real-world telemonitoring environments would provide valuable insights into operational performance outside controlled laboratory settings.

Overall, the proposed platform establishes a reproducible foundation for future IoMT research and demonstrates the potential of low-cost embedded technologies for supporting distributed physiological monitoring applications.

Supplementary Materials

Supplementary materials include the variable dictionary, metadata specification, repository documentation, example graph representations, and computational resources distributed through the public GitHub repository associated with this dataset.

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Conflicts of Interest

The authors declare no conflict of interest.

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